THISUL HERATH

ARTIFICIAL INTELLIGENCE UNDERGRADUATE

CONTACT

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PROFESSIONAL SUMMARY

IT Undergraduate specializing in Artificial Intelligence with a strong foundation in MLOps, machine learning engineering, and full-stack development. Proven track record of leading technical teams as a Team Lead on complex, multi-tier applications, including the 7 Super City enterprise retail platform and the UniVault mobile ecosystem. Experienced in building robust ML pipelines, containerized microservices, and secure, scaleable architectures using Python, Scikit-Learn, Node.js, and React Native. A disciplined leader, proven through serving as a Junior Prefect, House Captain, and multiple-time School and Provincial Colorsman for Badminton at Royal College. Passionate about engineering high-performance AI solutions and bridging the gap between data science and production-ready applications.

EDUCATION

2024 - Present — **BSc (Hons) in Information Technology Specializing in Artificial Intelligence (Current GPA - 3.84)**
SLIIT | 2024 – 2028

2023/24 — **GCE Advanced Level (A/L) - Physical Science Stream**
Royal College, Colombo 07.

2020/21 — **GCE Ordinary Level (O/L)**
Royal College ,Colombo 07.

PROJECTS

PYTHON | FASTAPI | SCIKIT-LEARN | MLFLOW | XGBoost | DOCKER | GITHUB ACTIONS
Link

Churn Prediction Service (Individual Project) | 2026

- Trained an XGBoost classifier on the Kaggle Telco Customer Churn dataset (7,043 records), tuned via a 30-trial Optuna Bayesian search optimizing for churn-class recall.
- Handled severe class imbalance using scale_pos_weight, achieving 87.2% recall on churners — prioritizing catching at-risk customers over raw accuracy, the business-correct trade-off for churn use cases.
- Benchmarked inference at 3.9ms for the full 1,409-row test set, confirming the model is fast enough for real-time serving.
- Built a FastAPI inference service with feature-alignment logic to guarantee consistent predictions between training and serving.
- Tracked all experiments, parameters, and model artifacts in MLflow for full reproducibility.
- Automated build, Pytest-based testing, and Docker image publishing through a GitHub Actions CI/CD pipeline.

REACT | TAILWIND CSS| NODE.JS | SUPABASE| PYTHON | SCIKIT-LEARN
Link

7 Super City: Intelligent Retail & Forecasting Platform (Group Project - Team Lead) | 2026

- Led a team building an enterprise retail platform combining POS, real-time inventory, loyalty analytics, and sales forecasting.
- Designed a dual-engine architecture separating transactional workloads (Express.js API) from ML workloads (Python microservice) for performance isolation.
- Implemented Row-Level Security across 15+ Supabase (PostgreSQL) tables and RBAC across 4 distinct portals using JWT and Bcrypt.
- Deployed Scikit-Learn time-series models to power AI-driven sales forecasting for inventory planning.

REACT NATIVE | EXPO | TYPESCRIPT | NODE.JS
Link

UniVault: Academic Collaborative Resource Platform (Group Project - Team Lead) | 2026

- Led development of a decentralized academic repository app for secure multi-format document sharing and subject-based study groups.
- Built a cross-platform mobile frontend with Expo Router for clean, file-based navigation.
- Designed a 7-model MongoDB schema with compound text indexing and schema hooks, backed by a Multer-based upload pipeline for both images and documents.
- Secured the platform with JWT authentication via React Context API and global error-handling middleware.

PYTHON | SCIKIT-LEARN NLTK | JUPYTER
Link

Automated NLP Text Preprocessing & Classification Pipeline (Group Project - Team Lead) | 2025

- Built a reproducible NLP pipeline: text preprocessing → TF-IDF vectorization → PCA dimensionality reduction → classification.
- Mitigated class imbalance with RandomOverSampler and tuned SVM/LinearSVC models via GridSearchCV for optimal performance.

JAVA | SPRING BOOT | MYSQL | BOOTSTRAP | JSP | MAVEN
Link

Wildlife Safari Trip Management System (Group Project - Team Lead) | 2025

- Led development of a full-stack platform digitalizing safari bookings, resource allocation, and coordination for 6 stakeholder roles.
- Built with Spring Boot (MVC, REST, Hibernate/JPA), including an automated vehicle/guide scheduling engine and a transactional finance & invoicing module.

CERTIFICATES

Certificate in AI/ML Engineer - SLIIT

- Stage 1 - https://code.sliit.org/certificates/nddgq8tgtd
- Stage 2 - https://code.sliit.org/certificates/cbxmlxyplk

SKILLS

- Languages: Python, JavaScript, TypeScript, Java, SQL
- AI / Machine Learning: Scikit-Learn, MLflow, NLTK, Imbalanced-Learn, Time-Series Forecasting, NLP Pipelines
- Frameworks & Libraries: React Native, React.js (Vite), Next.js, Node.js, Express.js, FastAPI, Spring Boot (MVC, REST)
- Databases & Cloud: MongoDB (Mongoose), Supabase (PostgreSQL), Firebase (Firestore), MySQL, Hibernate/JPA
- Tools & DevOps: Docker, GitHub Actions (CI/CD), Git, Expo / Expo Router, Jest, Pytest, Postman, Maven, Tailwind CSS, Figma

EXTRA-CURRICULAR ACTIVITY

Athletic Achievements & Colors

- Western Province Colors: Awarded for Badminton (2022, 2023) in recognition of provincial-level excellence
- School Colors (Royal College): Triple recipient of School Colors for Badminton (2019, 2022, 2023).
- House Captain (Badminton): Appointed as House Captain in 2023, overseeing team strategy and player development
- Vice-Captain : Served as Vice Captain of the Royal College Under-15 and Under-17 Badminton team (2018, 2020).

Leadership & Service

- Junior Prefect: Appointed as a Junior Prefect at Royal College (2018), responsible for maintaining student discipline and assisting in school-wide event management.

REFERENCES

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